

Omkar Chittar

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WORK EXPERIENCE

NBCUniversal

Orlando, FL

Lead Software Engineer

Nov 2025 – Present

- Independently designed and built a **full-stack estimating and takeoff platform** (Python backend, **React/TypeScript** frontend): document-ingestion pipeline, OCR, spatial registration, and a geometry-aware computation engine with AI-assisted extraction; cut takeoff time from **6 hours to 30 minutes**, backed by **~900 automated tests**
- Led software delivery across **Universal Creative** and **UDX**, partnering with stakeholders across **6 business functions** to gather requirements and author PRDs, shipping **10 products** from concept into production
- Architected **8+ production systems** spanning estimating, legal, engineering, safety, finance, and operations — designing shared patterns for service routing, validation, access control, and **human-in-the-loop approval**
- Led development of a centralized **contract lifecycle platform**, building ETL and a unified **search layer over 10,000+ documents** from SharePoint, databases, and enterprise repositories; reduced first-draft contract generation from ~2 weeks to **under 48 hours**
- Built retrieval and evaluation infrastructure over 5,000+ pages of legal documents with semantic search, automated citation validation, and deterministic checks; achieved full citation coverage at a **~2% review-error rate**
- Delivered a **cost-intelligence** platform over historical cost data with structured retrieval, query routing, provenance tracking, and a conversational UI; reduced cost-research time **95%**

FOX Sports

Redmond, WA (Remote)

AI Engineer

Aug 2024 – Nov 2025

- Decreased **issue resolution time 35%** by building an event-driven **monitoring service** (AWS Bedrock, Lambda, Slack API) for ML pipelines, surfacing drift alerts, exceptions, and health insights in real time
- Reduced **ad-hoc reporting effort 40%** by building a **natural-language-to-SQL** assistant (FAISS, Bedrock) serving producers at **p95 ~1.3s** latency
- Built an automated **evaluation and regression harness** (synthetic queries, oracle sets, precision@k) integrated into the iteration workflow; improved top-5 retrieval by **+12 p.p.**
- Cut per-request serving cost **28%** without quality loss via response caching and lightweight request routing on Databricks
- Streamlined operations on **2M+ digital assets** by automating metadata extraction and **ETL** (BigQuery, Spark, Cloud Functions); features powered engagement forecasting (LSTM, transformers) that lifted **user retention 25%**

Sakar Robotics

Pune, India

Software Engineer

June 2019 – June 2022

- Developed reusable **computer-vision** and **3D-processing modules** using OpenCV, Open3D, stereo depth, ICP, and multi-camera calibration; improved robotic pick success **18%** and reduced pose error **22%**
- Implemented TSDF mapping & 6-DoF tracking services, lowering re-localization time during occlusions by **35%**
- Developed **reinforcement-learning** components for robotic manipulation using **DDPG + HER**, improving task precision by **30%**

SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C++

Backend & Web: Node.js, Flask, REST APIs, React, Next.js

Cloud & Data: AWS (Bedrock, Lambda, SageMaker, S3), GCP (BigQuery, Vertex AI, Cloud Functions), Azure, Databricks, Snowflake, Spark, Airflow, Kafka, Postgres

MLOps/DevOps: MLflow, Docker, CI/CD, Git/GitHub, Linux, vector stores & databases, automated testing

GenAI/LLMs: RAG, agentic AI & tool calling, vector search, embeddings, guardrails, LLM evaluation (Ragas), prompt engineering & caching, LangChain, LangGraph, OpenAI, Claude, Hugging Face, MCP

ML/DL: Deep Learning, NLP, Computer Vision, PyTorch, TensorFlow

PUBLICATIONS

Two peer-reviewed papers on exoskeleton systems in *Materials Today: Proceedings*: [paper 1](#), [paper 2](#)

EDUCATION

University of Maryland

College Park, MD

Master of Engineering in Robotics — 3.97 CGPA

Aug 2022 – May 2024